# HPCVL and Compute Canada Workshop
# October/November 2010:

## Parallel Programming Using OpenMP, MPI and Posix Threads

**HPCVL Training Room, 115-993 Princess St., Kingston, Ontario**
October 25, November 1 & 8, 2010

Location and Time of Training Sessions:

Monday, October 25, 2010 (OpenMP) and
Monday, November 1, 2010 (MPI) and
Monday, November 8, 2010 (Posix Threads)

Nationally via Video Conferencing
Locally at the HPCVL Training Room in Kingston, Ontario.

Each Day: Registration: 8:30 am, Lectures: 9:00 am - 4:00 pm

Lecturers:
Hartmut Schmider (hartmut.schmider@queensu.ca) and Gang Liu (gang.liu@queensu.ca)
HPCVL, Queen's University, Kingston, Ontario

This workshop will be made available through *Compute Canada* by broadcasting via Video Conferencing from our Training Room in Kingston. If you would like to participate remotely, please contact your local High-Performance Consortium or university. If you are unsure how to get access, contact
**Ashley Wightman <ashley.wightman@queensu.ca> (613) 533-2561**

Please fill out Registration Form to sign up for the course
http://www.hpcvl.org/training/course_reg.html

Check the workshop webpage http://www.hpcvl.org/training/index.html for further details, such as the location of video conferencing rooms that are operated by HPCVL.

# Workshop Overview:

***Introduction to OpenMP***

This workshop introduces the OpenMP compiler directives to scientists who are interested in writing programs for shared-memory parallel computers, or who want to convert existing serial code to parallel. No previous knowledge about parallel programming is required, but we assume some basic background in programming , preferably in Fortran or C.
The use of OpenMP has become the *de facto* industry standard for parallel programming on shared-memory machines. It also makes implicit use of the multi-core nature of current CPU's. Most of the examples in this course are in Fortran, but the C programming language is considered as well. The workshop includes practical demonstrations.
Here is a short outline of the contents:

- Introduction to parallel programming, especially for shared memory
- OpenMP compiler directives
- Issues in shared-memory programming and how to resolve them
- Loop parallelism
- Explicit parallel regions
- Synchronization
- New features of OpenMP 3.0

***Introduction to MPI***

The *Introduction to MPI* workshop is directed at programmers and scientists, with a basic background in programming, who want to acquire basic skills in "parallelizing" code for a variety of platforms ranging from multi-processor servers to clusters. No prior knowledge of MPI or other message-passing systems is required. However, some background in programming in Fortran, C, or other languages would be helpful. The lectures include practical demonstration using an HPCVL Cluster. The following subjects will be addressed:

- MPI Basics: Programming Environments, Data Types, Communication
- Runtime Environments
- Parallel Principles and Programming Steps
- Combination of MPI with OpenMP
- Parallel Scheduling
- User-Defined Data Types
- All examples are in Fortran, C and C++

## Introduction to Posix Multithreading

The *Introduction to Posix Multithreading* workshop is for programmers and scientists with a basic background in C programming, who want to increase the flexibility and responsiveness of their code and take advantage of modern multi-core and multi-threaded computer architectures. This is a basic introduction to the Posix Thread Library and its application to the parallelization of C programs. We assume no prior knowledge of multithreading or parallel programming, but some background in Unix operating systems and programming in C will be necessary. All examples are in C. The lectures include demonstrations on a multicore machine. The following subjects will be addressed:

- Parallel Programming and Multithreading
- The Posix Thread Library, Basics of Thread Programming
- Creating and Manipulating of Threads
- Attributes
- Synchronization, Locks and Condition Variables
- Thread-Specific Data and Destructors